Ingest Data with Dataflows Gen2 in Microsoft Fabric

Core Definition and Purpose

- **Definition**: Dataflows Gen2 are cloud-based **ETL** (Extract, Transform, Load) tools used for building scalable data transformation processes.
- **Primary Function**: They enable users to ingest data from multiple sources, apply transformations, and land the cleansed data into a specified destination.
- **Low-Code Interface**: They utilize **Power Query Online**, providing a visual, low-to-no-code interface that allows transformations without writing complex scripts.

Key Features and Capabilities

- **Connectivity**: Supports a wide variety of connectors, including cloud/on-premises databases, Excel, SharePoint, Salesforce, and Fabric lakehouses.
- **Transformation Tools**: Offers diverse operations such as filtering, sorting, pivoting/unpivoting, merging/appending queries, and removing duplicates.
- **Data Destinations**: Transformed data can be loaded directly into Fabric destinations like a **Lakehouse, Warehouse, or SQL database**, as well as Azure-specific targets like Azure SQL or Azure Synapse Analytics.
- **Advanced View**: While visual-first, users can also view and edit the underlying **M code** through the Advanced Editor.

Integration and Orchestration

- **Pipeline Incorporation**: Dataflows can be included as an activity within **Data Pipelines**.
- **Orchestration**: Combining them with pipelines allows for complex workflows, such as executing scripts or stored procedures specifically after a dataflow finishes.
- **Automation**: Pipelines enable dataflows to be scheduled or triggered, removing the need for manual refreshes.

Benefits and Usage Scenarios

- **Efficiency**: Reduces data preparation time and prevents errors associated with manual extraction.
- **Reusability**: Promotes reusable ETL logic, which reduces the need for creating multiple connections to the same data source.
- **Analyst Support**: Allows data analysts to perform transformations closer to the source, supporting more efficient Power BI report development.
- **Consistency**: Ensures data quality by enabling standardizations, such as a consistent date dimension table across the enterprise.

Limitations

- **No Replacement**: Dataflows are not intended to replace a full data warehouse.
- **Security**: They do not support **row-level security**.
- **Licensing**: They require a **Fabric capacity workspace** to function.

Questions

1. While monitoring a Data Pipeline, you observe that a Dataflow Gen2 is not producing the expected output. What could be a possible reason for this issue?
A) The Power BI report is not connected to the dataflow.
B) The data pipeline trigger is configured for manual execution.
C) Incorrect transformation logic applied in the Dataflow Gen2.
D) The Lakehouse has reached its maximum storage capacity.

2. Your organization needs to automate the process of ingesting, transforming, and loading data into a Fabric lakehouse. Which combination of tools would you use to achieve this goal efficiently?
A) Only Data Pipelines.
B) Only Dataflows Gen2.
C) Dataflows Gen2 and Data Pipelines.
D) Power BI Desktop and Real-Time Hub.

3. Your company wants to use Dataflows Gen2 to transform and load data from various sources into a lakehouse. What is a key advantage of using Dataflows Gen2 for this task?
A) Dataflows Gen2 provide out-of-the-box row-level security for data sources.
B) Dataflows Gen2 automatically generate machine learning models.
C) Dataflows Gen2 can create a reusable semantic model for data analysts.
D) Dataflows Gen2 allow for the use of Python scripts within the transformation UI.

4. Your company needs to ensure data consistency and quality before loading it into your analytics platform. How does Dataflows Gen2 support this requirement?
A) By providing built-in machine learning algorithms to predict data quality issues.
B) By automatically applying row-level security during data transformation.
C) By enabling users to clean and transform data before loading it to a destination.
D) By restricting access to the Lakehouse until data passes a validation test.

5. Your organization wants to expose data to a large group of data analysts without making the data source complexity apparent. How can Dataflows Gen2 assist in this situation?
A) By simplifying data source complexity and exposing only the dataflows to the analysts.
B) By implementing row-level security to restrict data access based on user roles.
C) By automatically updating data models using artificial intelligence.
D) By migrating all analysts to a dedicated Spark cluster.

6. Which component is often overlooked but is essential for automating data refresh in a Dataflow Gen2?
A) Scheduling the dataflow in a pipeline.
B) Applying data masking on sensitive columns.
C) Integrating the dataflow with Power BI Desktop.
D) Creating a manual shortcut in OneLake.

7. In what scenario is Dataflows Gen2 preferred over traditional data pipelines for data ingestion and transformation?
A) When the goal is to replace an existing data warehouse entirely.
B) When row-level security is a critical requirement for the data processing.
C) When you need to integrate data from multiple sources with complex transformations using a low-code interface.
D) When the data is already in Parquet format and requires no changes.

8. Which feature of Dataflows Gen2 enhances data integration by reducing complexity for larger analyst groups?
A) Direct integration with Power Apps.
B) Exposing only dataflows instead of raw data sources.
C) Manual execution of data refreshes via a command-line interface.
D) Automatic generation of SQL stored procedures.

9. In a Dataflow Gen2 process, what component is critical to ensure that transformed data is available for downstream analytics?
A) Incorporation of machine learning models within the dataflow.
B) A designated data destination for loading transformed data.
C) Automatic integration with Power BI dashboards.
D) The presence of a head node in the Spark pool.

10. Which interface does Dataflow Gen2 use to provide the low-code transformation experience?
A) Visual Studio Code.
B) Power Query.
C) SQL Server Management Studio.
D) Jupyter Notebooks.

11. What is a "Data Destination" in the context of Dataflow Gen2?
A) The original source where data is stored.
B) A folder in OneLake used only for temporary staging.
C) The target system (like a Lakehouse or Warehouse) where transformed data is loaded.
D) A Power BI report that consumes the final data.

12. Which activity in a Data Pipeline is used to execute a Dataflow Gen2 as part of an orchestrated sequence?
A) Copy Data activity.
B) Notebook activity.
C) Dataflow activity.
D) Stored Procedure activity.

13. What happens during the "Publish" phase of a Dataflow Gen2?
A) The data is immediately sent to a public website.
B) The metadata and transformation logic are saved and validated.
C) A Power BI dashboard is automatically created.
D) The user is charged for the total storage used in the Lakehouse.

14. Which storage technology serves as the foundation for all data storage in Microsoft Fabric, including Dataflow Gen2 outputs?
A) Azure SQL Database.
B) OneLake.
C) Amazon S3.
D) Local SSD storage.

15. If you need to perform high-scale data movement without any transformations, which Fabric component is most efficient?
A) Dataflow Gen2.
B) Data Pipeline (Copy activity).
C) Power BI Paginated Reports.
D) Fabric Activator.

16. How does Dataflow Gen2 handle large-scale transformations differently than the original Power BI Dataflows?
A) It only supports CSV files.
B) It leverages the Fabric compute engine to load data to supported destinations.
C) It requires a manual gateway for all cloud connections.
D) It cannot be scheduled.

17. Which of the following is NOT a supported data destination for Dataflow Gen2?
A) Fabric Lakehouse.
B) Fabric Warehouse.
C) Azure SQL Database.
D) Local Excel file on a user's desktop.

18. Why would a developer use a Data Pipeline to "orchestrate" multiple Dataflows?
A) To ensure Dataflows run in a specific order with dependency logic.
B) To convert Dataflows into Python code automatically.
C) To reduce the cost of OneLake storage.
D) To bypass the need for a Fabric capacity.

19. What is the primary benefit of the "Low-code" nature of Dataflow Gen2?
A) It makes the data processing slower but more secure.
B) It allows non-developers and analysts to build complex ETL processes visually.
C) It requires the use of advanced SQL syntax for every step.
D) It limits the number of data sources that can be connected.

Answers

1. C
2. C
3. C
4. C
5. A
6. A
7. C
8. B
9. B
10. B
11. C
12. C
13. B
14. B
15. B
16. B
17. D
18. A
19. B